INCH-POUND

MIL-C-39019/2F
26 June 1989
SUPERSEDING
MIL-C-39019/2E
4 October 1982

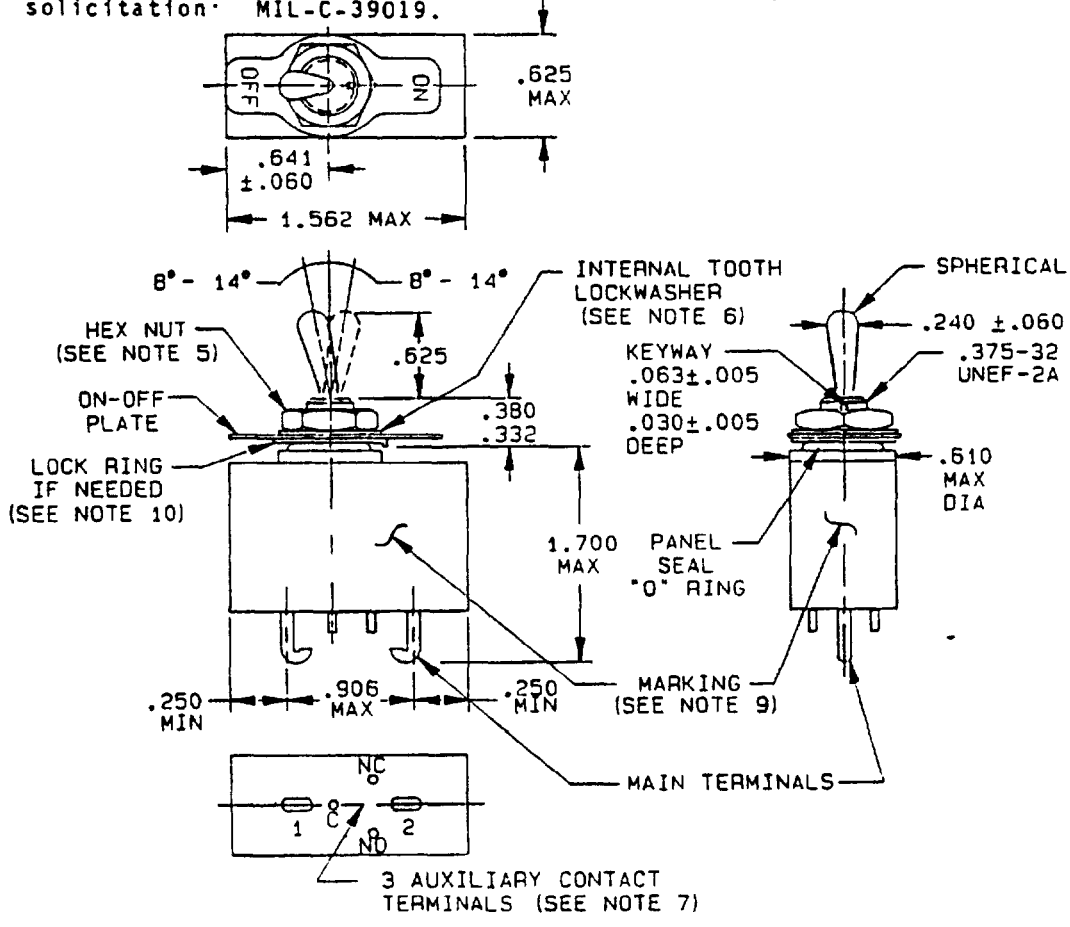
MILITARY SPECIFICATION SHEET

CIRCUIT BREAKERS, MAGNETIC, LOW-POWER, SEALED, TRIP-FREE

ONE-POLE, AUXILIARY CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39019.



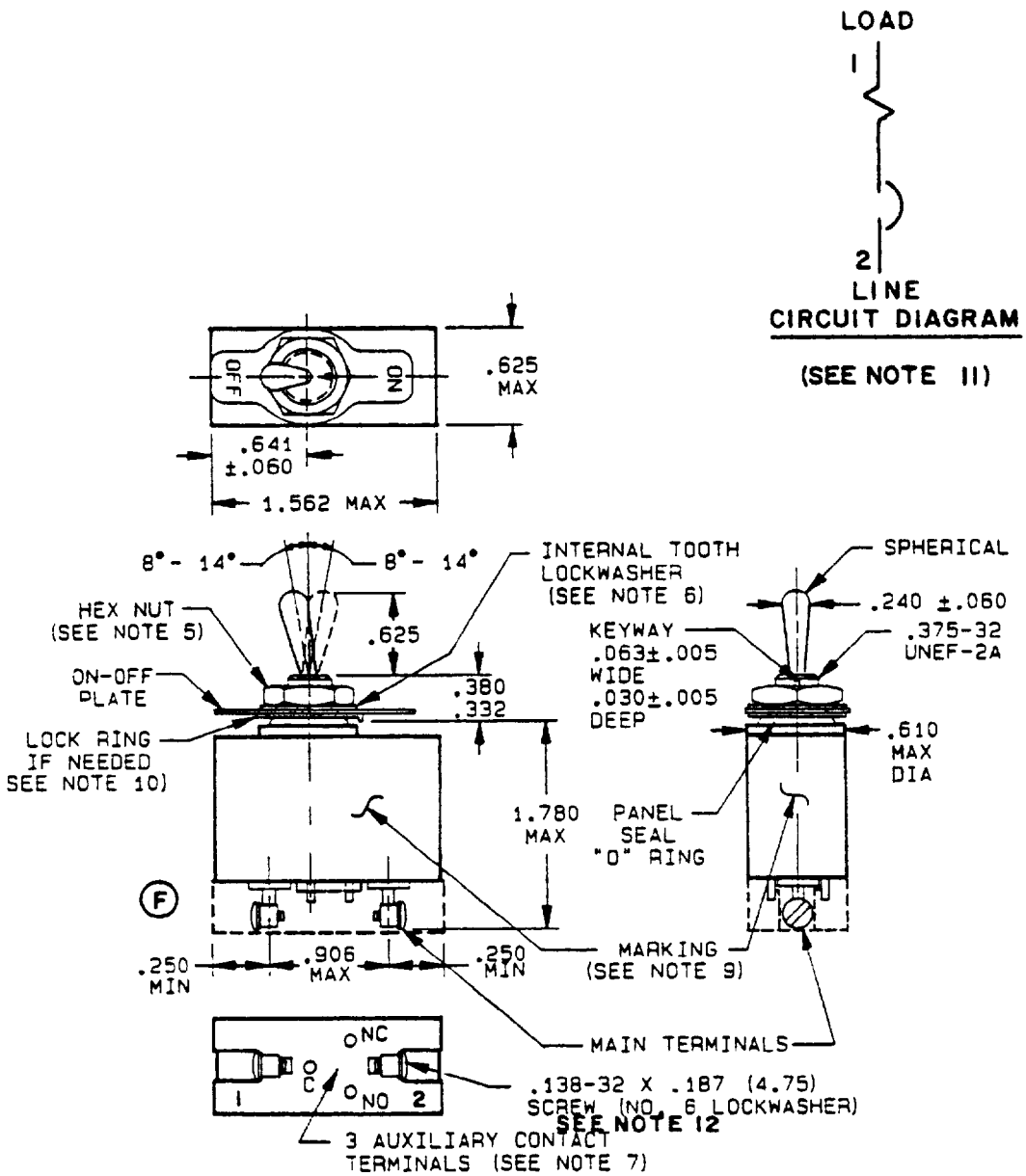
CONFIGURATION 1

FIGURE 1. Circuit breaker configuration.

(F) denotes changes

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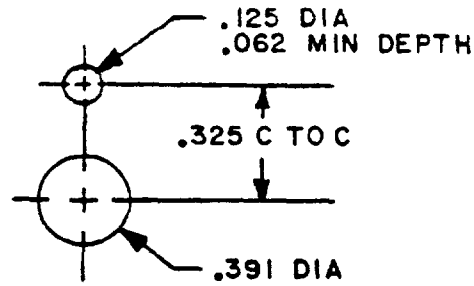
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CONFIGURATION 2

FIGURE 1. Circuit breaker configuration - Continued.

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PANEL MOUNTING DETAIL

Inches	mm	Inches	mm
.005	0.13	.382	9.70
.030	0.76	.391	9.93
.060	1.52	.392	9.96
.062	1.57	.493	12.52
.063	1.60	.507	12.88
.125	3.18	.610	15.49
.138	3.51	.625	15.88
.240	6.10	.641	16.28
.250	6.35	.906	23.01
.325	8.26	1.562	39.67
.332	8.43	1.700	43.18
.375	9.52	1.780	45.21
.380	9.65		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Envelope design optional.
(F) 5. Hex mounting nut .375-32 UNEF-2B thread, .093 ±.005 (2.36 ±0.13 mm) thick or equivalent MS25082-20, -B20.
(F) 6. Internal tooth lockwasher, .507/.493 O.D., .392/.382 I.D., .025 ±.005 (0.64 ±0.13 mm) thick, stainless steel or equivalent MS35333-134.
7. Auxilliary contact terminals shall be located on the back side of the circuit breaker (same surface as main terminals), but otherwise their location, style and design shall be optional, consistent with dimensional or other specified requirements. Auxilliary contact terminals shall accommodate two number 20 AWG wires.
8. The effective bushing thread length (not including mounting hardware) is .280 (7.11 mm) minimum.
9. Marking may appear on any surface except the mounting surface.
10. Lock ring not required if on-off plate has locking tab.
11. Numerical marking optional on circuit diagram.
(F) 12. For configuration 2 add "S" at end of dash number. Main terminal screw MS35265-25 or equivalent. Lockwasher, flat-internal tooth, number 6 or MS35333-71.

FIGURE 1. Circuit breaker configuration - Continued.

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REQUIREMENTS:

Ⓕ Dimensions and configuration See figure 1 and note 12.

Weight. For reference only, approximately 50 grams.

Mounting hardware. See figure 1.

Terminals: See figure 1.

Voltage and frequency rating: 50 V dc, maximum and 240 V ac, maximum, at 60 and 400 Hz.

High inrush: 800 percent at 60 Hz and dc, 1400 percent at 400 Hz (see table I).

Current rating: See table I.

Circuit diagram: See figure 1.

Time delay: See table I.

Resistance or impedance: See table I.

Ⓕ Part or Identifying Number (PIN): The PIN shall be M39019/02-(dash number taken from table I) (e.g., M39019/02-200 or M39019/02-200S).

Cross reference: See table II.

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Ⓕ TABLE I. Circuit breaker dash numbers and applicable characteristics.

Dash number 1/		Current rating (amps)	Time delay	Resistance or impedance (ohms - max) 2/		
Standard delay	High inrush			DC	60 Hz	400 Hz
200	300	0.05	A	680.0	690.0	710.0
201	301	0.05	B	680.0	690.0	710.0
202	---	0.05	C	680.0	690.0	710.0
203	302	0.1	A	150.0	170.0	180.0
204	303	0.1	B	150.0	170.0	180.0
205	---	0.1	C	150.0	170.0	180.0
206	304	0.25	A	20.0	26.0	27.0
207	305	0.25	B	20.0	26.0	27.0
208	---	0.25	C	20.0	26.0	27.0
209	306	0.5	A	5.4	6.0	6.6
210	307	0.5	B	5.4	6.0	6.6
211	---	0.5	C	5.4	6.0	6.6
212	308	0.75	A	2.5	2.7	2.8
213	309	0.75	B	2.5	2.7	2.8
214	---	0.75	C	2.5	2.7	2.8
215	310	1.0	A	1.35	1.5	1.61
216	311	1.0	B	1.35	1.5	1.61
217	---	1.0	C	1.35	1.5	1.61
218	312	2.0	A	0.40	0.40	0.50
219	313	2.0	B	0.40	0.40	0.50
220	---	2.0	C	0.40	0.40	0.50
221	314	2.5	A	0.25	0.25	0.27
222	315	2.5	B	0.25	0.25	0.27
223	---	2.5	C	0.25	0.25	0.27
224	316	3.0	A	0.15	0.15	0.17
225	317	3.0	B	0.15	0.15	0.17
226	---	3.0	C	0.15	0.15	0.17
227	318	4.0	A	0.10	0.10	0.12
228	319	4.0	B	0.10	0.10	0.12
229	---	4.0	C	0.10	0.10	0.12
230	320	5.0	A	0.061	0.063	0.072
231	321	5.0	B	0.061	0.063	0.072
232	---	5.0	C	0.061	0.063	0.072
233	322	6.0	A	0.042	0.043	0.050
234	323	6.0	B	0.042	0.043	0.050
235	---	6.0	C	0.042	0.043	0.050
236	324	7.0	A	0.036	0.036	0.040
237	325	7.0	B	0.036	0.036	0.040
238	---	7.0	C	0.036	0.036	0.040
239	326	7.5	A	0.031	0.031	0.038
240	327	7.5	B	0.031	0.031	0.038
241	---	7.5	C	0.031	0.031	0.038
242	328	8.0	A	0.027	0.028	0.035
243	329	8.0	B	0.027	0.028	0.035
244	---	8.0	C	0.027	0.028	0.035
245	330	9.0	A	0.022	0.022	0.028
246	331	9.0	B	0.022	0.022	0.028

See footnotes at end of table.

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Ⓕ TABLE I. Circuit breaker dash numbers and applicable characteristics - Continued.

Dash number 1/		Current rating (amps)	Time delay	Resistance or impedance (ohms - max) 2/		
Standard delay	High inrush			DC	60 Hz	400 Hz
247	---	9.0	C	0.022	0.022	0.028
248	332	10.0	A	0.018	0.021	0.024
249	333	10.0	B	0.018	0.021	0.024
250	---	10.0	C	0.018	0.021	0.024
251	334	12.5	A	0.012	0.013	0.015
252	335	12.5	B	0.012	0.013	0.015
253	---	12.5	C	0.012	0.013	0.015
254	336	15.0	A	0.009	0.009	0.010
255	337	15.0	B	0.009	0.009	0.010
256	---	15.0	C	0.009	0.009	0.010
257	338	20.0	A	0.006	0.006	0.007
258	339	20.0	B	0.006	0.006	0.007
259	---	20.0	C	0.006	0.006	0.007
260	340	1.25	A	.9	1.0	1.1
261	341	1.25	B	.9	1.0	1.1
262	---	1.25	C	.9	1.0	1.1
263	342	1.5	A	.65	.70	.75
264	343	1.5	B	.65	.70	.75
265	---	1.5	C	.65	.70	.75

1/ For configuration 2, add "S" at end of dash number.

2/ The corresponding maximum wattage losses, which in no case shall exceed 3.0 watts, may be calculated as I^2R or I^2Z.

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TABLE II. Cross reference of superseded part numbers.

Superseded MIL-C-39019/2B part number M39019/2-	New part number M39019/02-	Superseded MIL-C-39019/2B part number M39019/2-	New part number M39019/02-	Superseded MIL-C-39019/2B part number M39019/2-	New part number M39019/02-
1	200	42	231	83	218
2	201	43	230	84	219
3	200	44	231	85	218
4	201	45	230	86	219
5	200	46	231	87	227
6	201	47	230	88	228
7	200	48	231	89	227
8	201	49	230	90	228
9	200	50	231	91	227
10	201	51	248	92	228
11	209	52	249	93	233
12	210	53	248	94	234
13	209	54	249	95	233
14	210	55	248	96	234
15	209	56	249	97	233
16	210	57	248	98	234
17	209	58	249	99	236
18	210	59	248	100	237
19	209	60	249	101	236
20	210	61	254	102	237
21	215	62	255	103	236
22	216	63	254	104	237
23	215	64	255	105	242
24	216	65	254	106	243
25	215	66	255	107	242
26	216	67	254	108	243
27	215	68	255	109	242
28	216	69	254	110	243
29	215	70	255	111	245
30	216	71	257	112	246
31	224	72	258	113	245
32	225	73	257	114	246
33	224	74	258	115	245
34	225	75	257	116	246
35	224	76	258	117	251
36	225	77	257	118	252
37	224	78	258	119	251
38	225	79	257	120	252
39	224	80	258	121	251
40	225	81	218	122	252
41	230	82	219		

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 99
DLA - ES

User activities:
Army - AV, AT
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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